HR Attrition Analysis

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BACKGROUND

A Data Analyst at XYZ.ltd has a job descripting data to provide insight about employees and management on how to implement HR strategy in order to have more productive employees.

The management has emailed to ask insight to understand what factor they should focus on, in order to reduce attrition. They want to know which of the variables is most important and needs to be addressed right away

IDENTIFY BUSSINESS INFORMATION

Company; XYZ ltd.

User: Chief Human Resource Officer

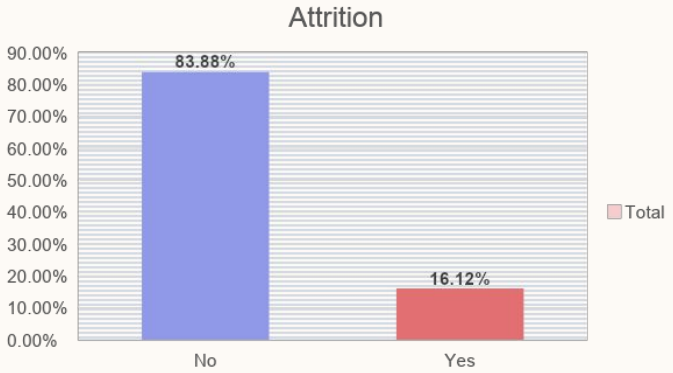
Assignment: To determine what factor that should be focus on in order to reduce attrition

Company with 4.000 Employee

Current attrition rate: 15%

Dataset Sample: 4.410 employee

Attrition	Count of ID
No	3.699
Yes	711
Grand Total	4.410



Dataset file:

- W1_employee_survey_data
- W1_general_data
- W1_manager_survey_data
- W1_in_time
- W1_out_time

STAKEHOLDER'S ROLE

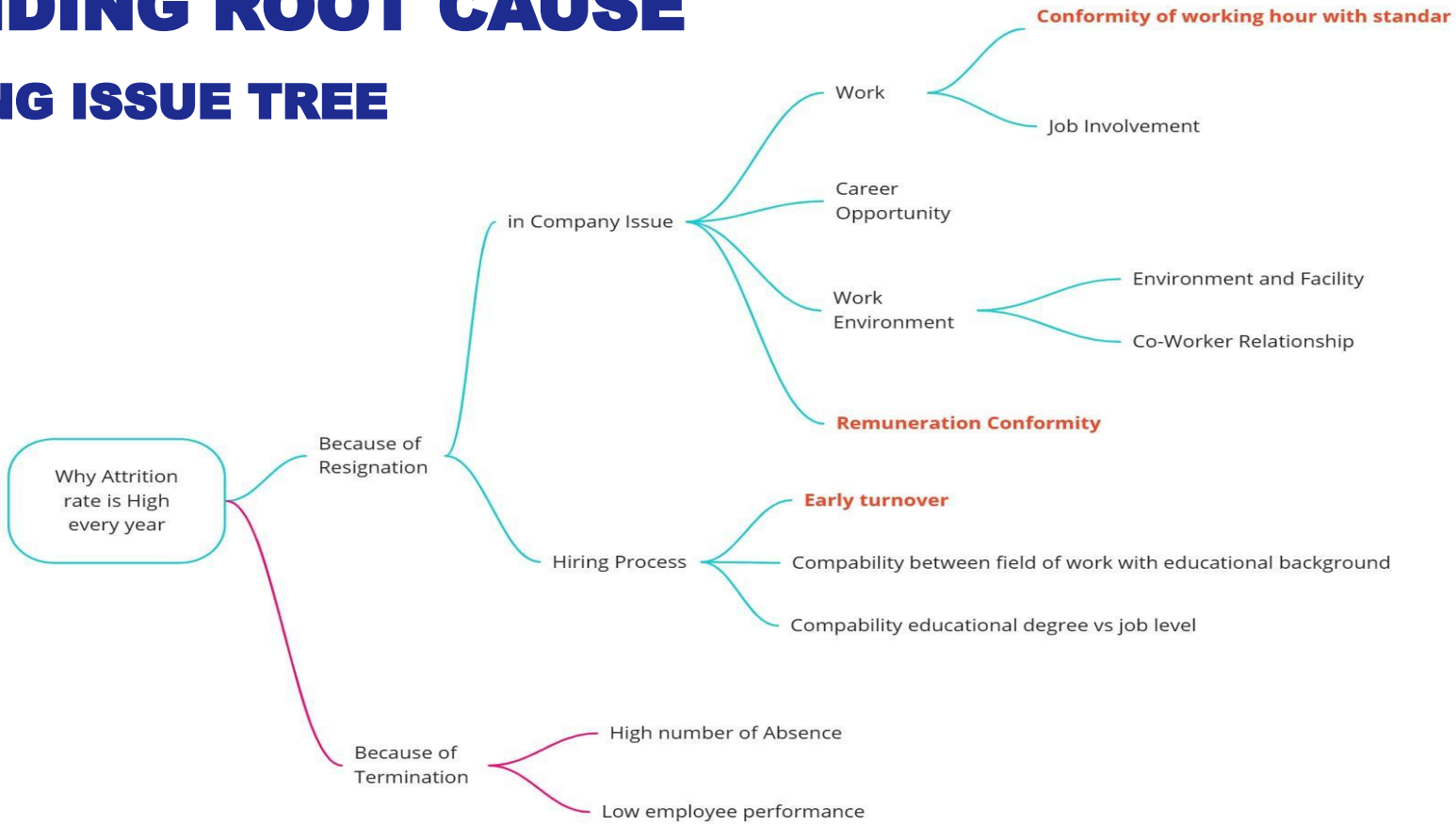
D	A	R	C	I
The ultimate decision maker	Person who is held solely responsible for whether or not the project gets done.	The person or persons literally responsible for "doing the thing" on a project.	Individuals who may be consulted or asked for their input.	Persons who need to be kept informed of a project's progress
CHRO	Head Of Data	Data Analyst HR Associate	Data Engineer Team HR Team	Business operations Team Product Team Business Development Team Marketing Team Enginering Team

PROBLEM & OBJECTIVES

Problem Statement	How to reduce the attrition ratio in the company from 15% to 10% within 12 Month.
Define Objective	To find an appropriate method to reduce the attrition ratio from 15% to 10% within 12 Month

FINDING ROOT CAUSE

USING ISSUE TREE



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PRIORITIZE HYPHOTHESIS

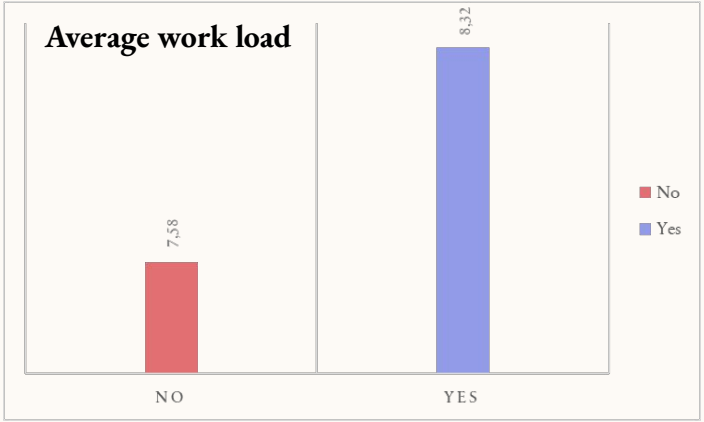
Possible Root Cause	Hypothesis	Prioritization
Comformity of Working Hours with Standards	If you adjust the number of working hours to the available standards, the attrition rate will decrease by 5% in a year	**High:** because overwork can cause stress and the producivity of eployment can decreese
Remuneration Comformity	If the amount of salary gap is less than 2% from market standards for a field of work then the attrition rate will decrease by 5% in a year	**High:** Employees are largely motivated to contribute when they are paid well
New Hire Turnover	If we reduce early turn over from 14,6% to 10% then the attrition rate wil decrease by 3% in a year	**High:** This is important data since recruiting and training new employees can be expensive.
Job Involvement	If there is no gap in the division of labor, then the attrition rate will decrease by 2% in a year	Mid
Career Opportunity	If it provides clear career opportunities with average training twice a year for new and talented employees then the attrition rate will decrease from 5% in a year	Mid
High number of Absence	if we lowering the absenteeism rate from 3,2% to 2,8% we can lower attrition rate by 2%	Mid
Environment & Facility	If employees get the comfort of the work environment well (average value of environment satisfaction 3.5) then the attrition rate will decrease by 2% in a year	Low
Co-Worker Relationship	If the relationship between co-worker is good (average relationship satisfaction 3.5) then the attrition rate will decrease by 2% in a year	Low
Compability Between Field of Work with Educational Background	If the field of work matches the educational background, the attrition rate will decrease by 2% in a year	Low
Compability educational degree vs	If the level of education is in accordance with the job level, the attrition rate will	Low
Low employee performance	If we increase the value of employee performance, the attrition rate will decrease by 2% in a year	Low

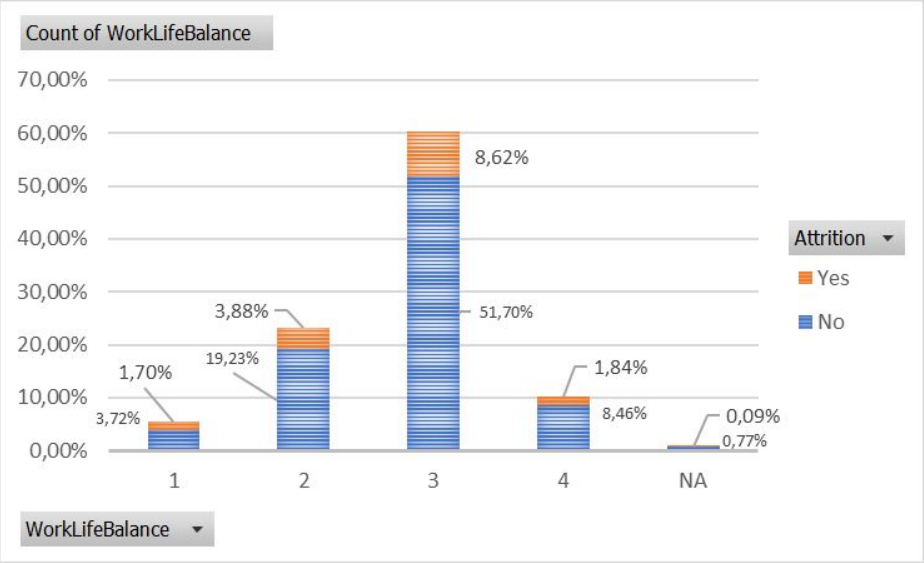
DEFINE METRICS FOR HIGH PRIORITY HYPOTESIS

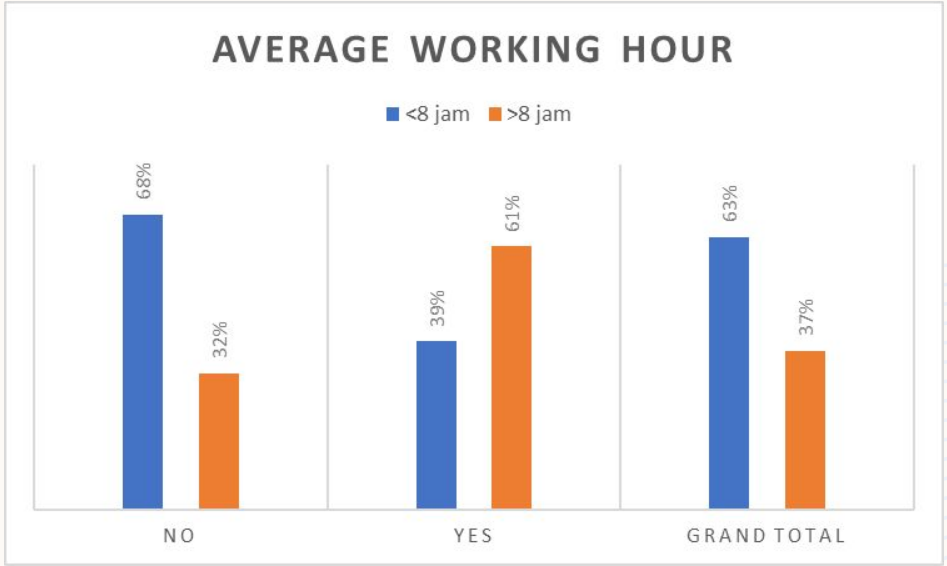
Possible Root Cause	Hypothesis	Prioritization	Metrics	Metric Reasoning
Comformity of Working Hours with Standards	If you adjust the number of working hours to the available standards, the attrition rate will decrease by 5% in a year	**High:** because overwork can cause stress and the producivity of eployment can decreese	**Overtime percentage =** (overtime pay amount / total payroll)	This metric can help the company to reveal if there are enough employees and whether managers are maintaining an appropriate schedule.
			% employee overtime sum of employee do overtime / total employee	To show the percentage of overtime employees
			Employee work live balance	To see is the employe able to manage their work life balance to avoid stress
Remuneration Comformity	If the amount of salary gap is less than 2% from market standards for a field of work then the attrition rate will decrease by 5% in a year	**High**: Employees are largely motivated to contribute when they are paid well	Employee base Salary Increase rate	To show the percentage increase in the base salary over a period of time.To show the percentage increase in the base salary over a period of time.
			Comparation Rate	compares an employee's salary with the market midpoint (middle of the range salary) for similar positions at other companies
New Hire Turnover	If we reduce early turn over from 14,6% to 10% then the attrition rate wil decrease by 3% in a year	**High:** This is important data since recruiting and training new employees can be expensive.	Early Turn Over Rate	Early turnover measures the percentage of new hirees leaving within the first year, and it may give sign whether there is room for improvement with onboarding or recruitment strategies.

TOP 3 HYPOTESIS

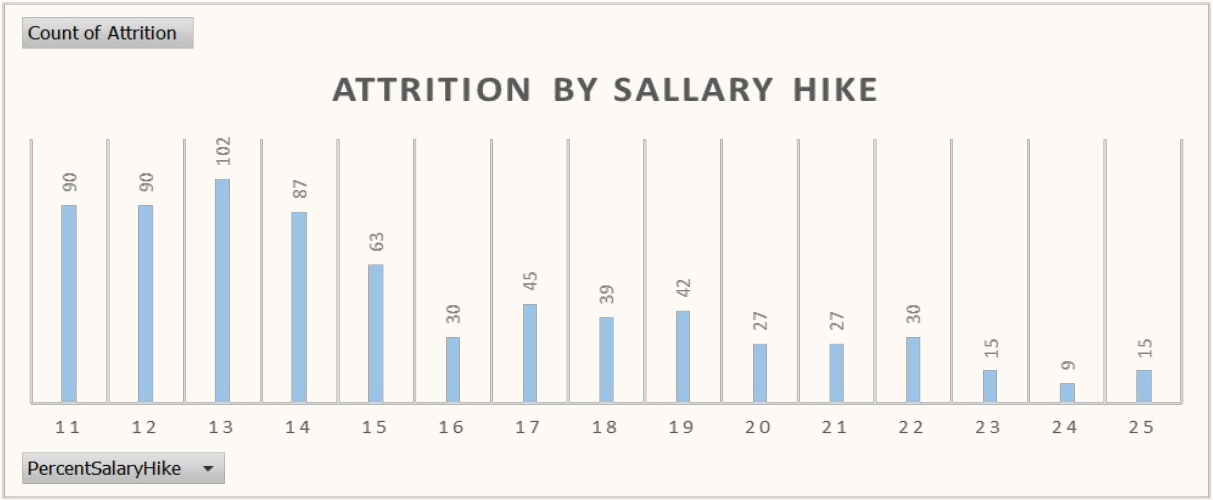
1. If you adjust the number of working hours to the available standards, the attrition rate will decrease by 5% in a year

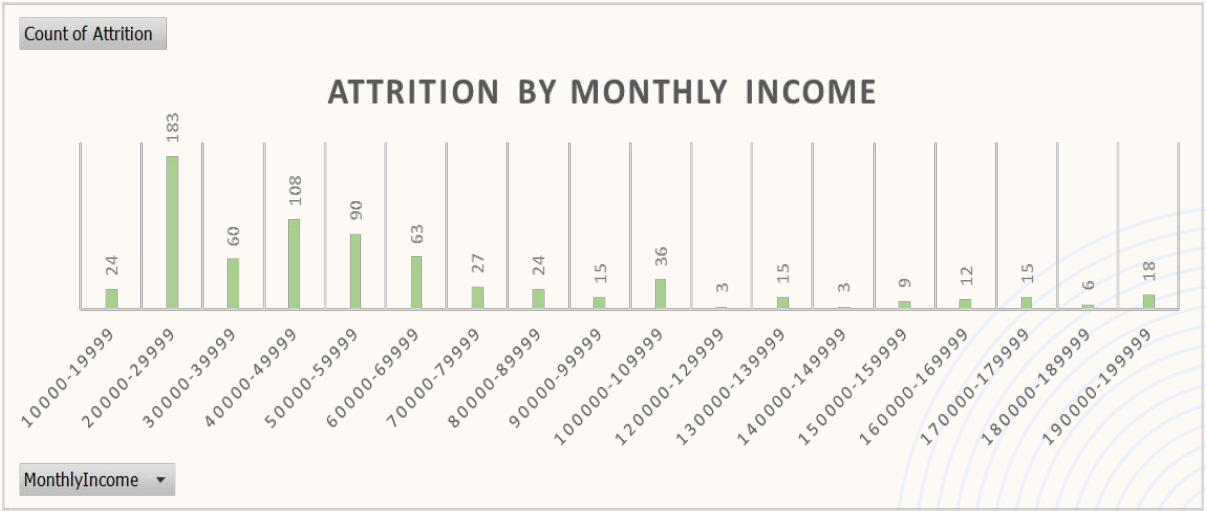




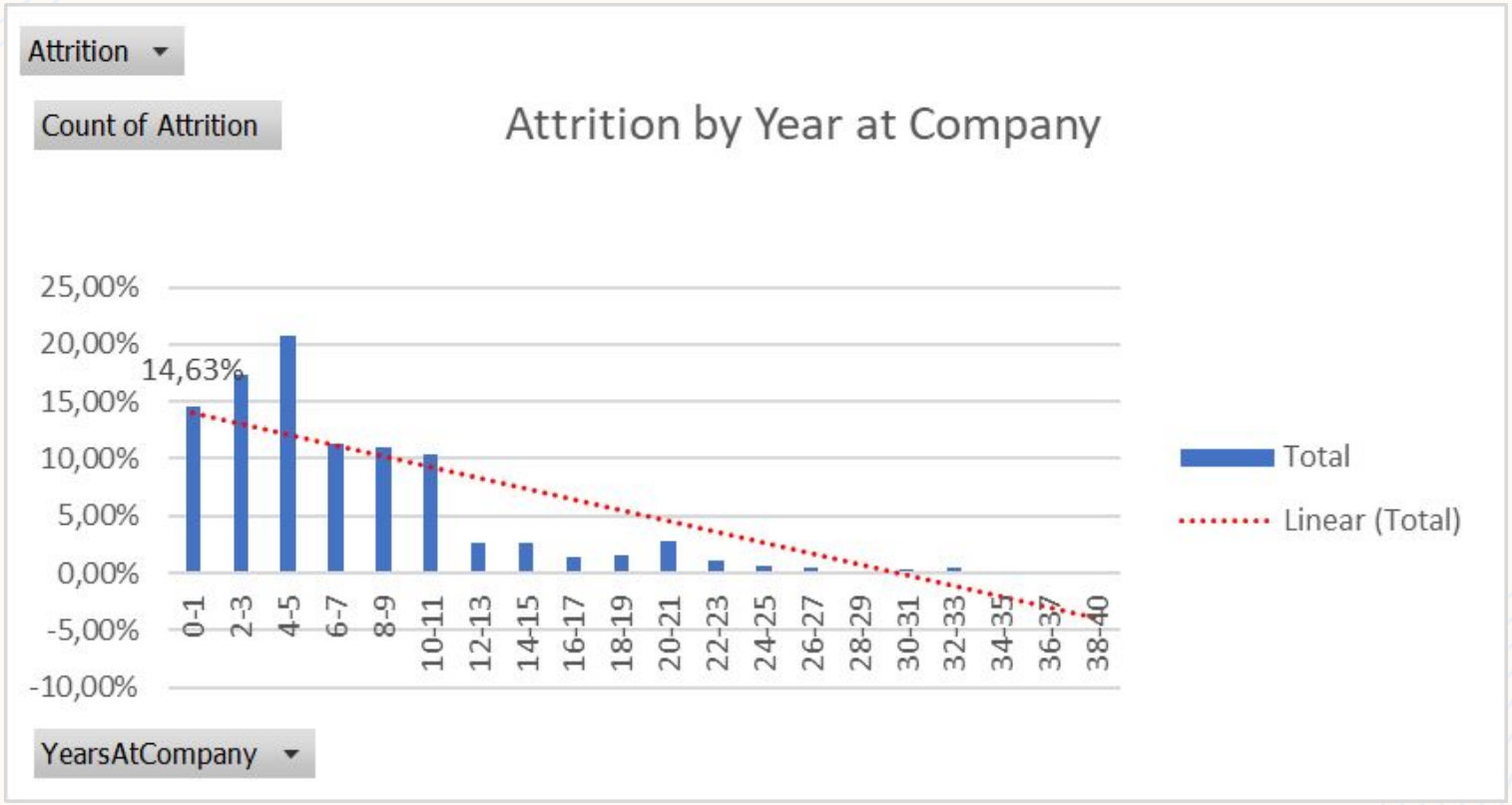


2. If the amount of salary gap is less than 2% from market standards for a field of work then the attrition rate will decrease by 5% in a year





3. If we reduce early turn over from 14,6% to 10% then the attrition rate wil decrease by 3% in a year



HR Attrition Dashboard

Overview

Employee	Attrition Rate
4.410	16.12 %

Attrition	Retention
711	3.699

Demographic

Attrition Retention Department (All)

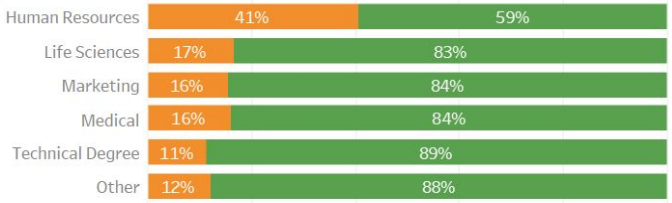
Gender







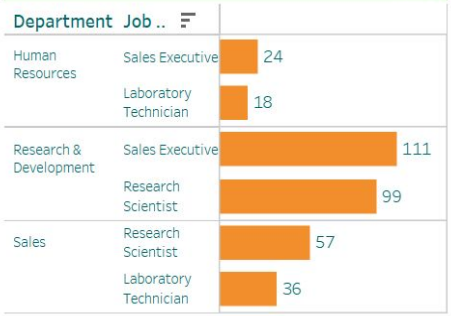
Educational Field



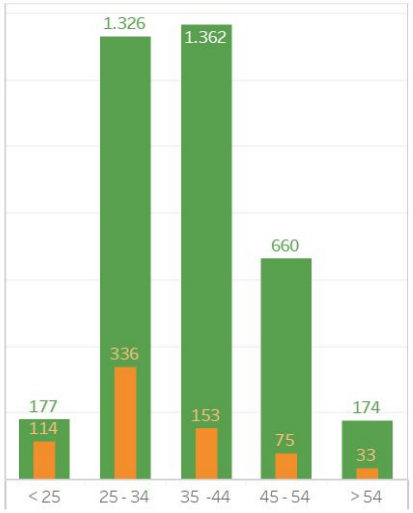
Departmen



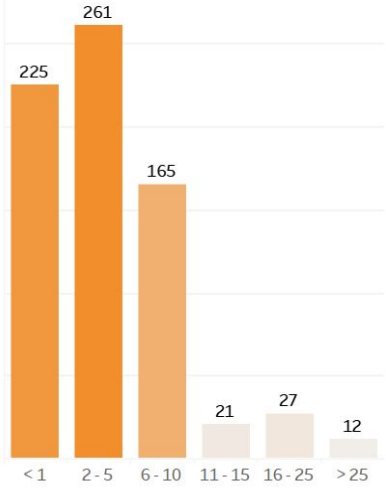
Job Role



Age



Years at Company



Survey

0 — 3.000

Work Life Balance

1	2	3	4
239	1.019	2.660	454

Environment Satisfaction

835	850	1.340	1.322

Job Satisfaction

853	834	1.312	1.353

Job Involvement

248	1.112	2.567	425

Performance Rating

238	1.016	2.646	452

THANK YOU